

from native plants, which affect alien species, but leave native ones unharmed. Killing for conservation is a difficult and divisive issue, particularly when the victims are cats that have run wild. However, there is no doubt—as far as Australian marsupials are concerned—that it is a highly effective measure.

Introduced species are even more of a problem on offshore islands, where they can devastate land animals and colonies of nesting birds. Many of the world's remotest islands, such as Kerguelen in the southern Indian Ocean, have been overrun by rats, which arrived aboard ships several centuries ago. Rats can be extremely difficult to control and on Kerguelen at least, eradication programs have not succeeded. However, on several islands off the coast of New Zealand, rats have been eradicated to create safe havens for tuataras—among the most endangered reptiles in the world. The small size of these islands makes them ideal “arks,” because they are relatively easy to keep alien-free.

Legal protection

A host of national laws and international agreements serve to protect wildlife and reduce the risk of species being driven toward extinction. The Convention on International Trade in Endangered Species (CITES) came into force in 1975, in order to curb trade of living plants and animals or their products. It protects 35,000 species, assigning each to one of three lists (the so-called Appendices), depending upon the level of threat and degree of protection needed. Other international bodies have more specific concerns. The International Whaling Commission (IWC) was set up in 1946 to oversee “sustainable harvesting” of whales, but—in the face of plummeting numbers—introduced a moratorium on commercial whaling in 1986. Countries that are party to international agreements, such as CITES, have domestic legislation to regulate at a national level. At the same time national legislation is used to help enforce conservation in protected areas, such as national parks. Nearly 15 percent of land and just over 1 percent of ocean areas are protected by restricting hunting and habitat clearing. Illegal activity, nevertheless, continues to frustrate efforts to conserve some habitats or species. Some wildlife—or their products—are

very valuable on the black market, fueling poaching and illicit trade. Since 1960, a 90 percent drop in numbers of black rhinoceroses is almost entirely due to poaching—making it critically endangered.

Commercial exploitation

Few people would condone the sale of rhino horn or tiger bones, but some conservationists do believe that—where possible—wild animals should be made to “pay their way”. According to this viewpoint, animals are best conserved if they generate income, as this provides an incentive for protecting them. There are 2 main ways by which this can happen; wildlife tourism can be encouraged, with some of the revenue being used for conservation work; alternatively, animals themselves can be managed as a resource.

Wildlife tourism is a booming business, although it has undeniable drawbacks, such as increasing habitat disturbance. But wildlife experts are often

TOURIST ATTRACTION

Watched by a group of tourists, a cheetah relaxes in the evening sunshine. Its tameness is unusual and is a sign that tourism's intrusion into its habitat is affecting its natural behavior.

sharply divided about the use of animals as a resource. In recent years, the African elephant has been a case in point, with different conservation bodies at odds about the exploitation of ivory. In this debate, one side believes that the legal sale of ivory is bound to have a damaging effect on elephant numbers. The other side believes that if it is carefully controlled, the sale of ivory could actually safeguard the species by generating money to protect it.

At present, no one knows whether commerce has a real place in wildlife conservation. If it does, one factor is certain: the income generated by wild animals will have to benefit local people, as they are the ones who can make conservation work.

CITES

The Convention on International Trade in Endangered Species (CITES) has over 120 signatories, and is the most important piece of international legislation governing the movement of live animals and animal products across international borders. CITES completely prohibits trade in over 400 species, and requires special permits for trade in others. Some illegally traded objects are easy to identify, but others can be distinguished by DNA analysis—a relatively new technique increasingly used by customs officials. CITES has been successful in some areas, but despite increasing vigilance at ports and airports, smuggling is still a problem.

BANNED GOODS

All the items shown in this photograph are made of turtle shell—an animal product that cannot be exported under CITES regulations. In some countries, national laws make it illegal to own animals or objects listed by CITES.



WINNERS AND LOSERS

Almost extinct in the 1930s, the Antarctic fur seal now numbers millions. Its recovery is owed not only to protection from hunting but also to reduced numbers of whales, which compete for krill.

